



Is the Dutch disease well and alive? A cross-country assessment of mining spillovers on employment in the 2002–2014 period

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ARTICLE INFO

Keywords:

Dutch disease
Mining
Extractive sectors
Employment spillovers
Elasticities

ABSTRACT

Following the Dutch disease model of Corden and Neary (1982), the expansion of a booming extractive sector crowds out employment in key sectors, i.e., manufacturing. To this day, these potential negative labor spillovers remain a cause of concern for policymakers. Empirical literature has explored this question by focusing on regional dynamics of employment, but country-wide effects remain understudied. This paper explores the mining sector's employment spillovers at the national level over the 2002–2014 period. Using data from WIOD, we employ Morretti's (2010) specification, as extended in Fleming and Measham (2014), and adopt a GMM model to address potential endogeneity issues. Following the decomposition in Foster-McGregor (2019), we differentiate mining employment expansion based on a) changes in domestic demand and b) changes due to exports. Along with regional studies, our results indicate that there is no evidence of negative spillovers on manufacturing jobs, regardless of the source of mining jobs expansion (domestic or exports). Such results hold even in countries with relatively low levels of manufacturing output. We do find some evidence of negative effects in agriculture, and positive effects on construction and services; this, however, seems to hold only in countries with relatively low manufacturing output. Findings also suggest that positive elasticities at the national level, when observed, are quite moderate: e.g., a 10% increment in mining employment leads to a 1.1%–1.4% increment in construction employment.

1. Introduction

Understanding the indirect effects, that extractive industries have on employment in other sectors at different scales has a two-fold motivation. Firstly, it allows for the assessment of the overall economic impacts of mining, i.e., oil, gas, and mineral operations. The mining industry is well-known for its capital-intensive nature limiting direct employment opportunities, thus, understanding the magnitude of employment generation in other sectors sector is essential for policy makers. Measuring spillovers becomes even more relevant as mining operations move towards higher levels of automation which further reduce direct employment opportunities – a trend that will have stronger adverse effects for low-skilled jobs in developing countries (Cosbey et al., 2016; Leeuw and Mtegha, 2018).

This question has been particularly addressed at the regional level as many of the environmental and social costs related to the exploitation of extractive resources are locally observed. The assessment of positive spillovers of mining employment, consequently, is highly linked to social licensing and the relation with the local community where it takes place. Most empirical studies have addressed this matter by calculating indirect employment effects based on input-output methodologies (e.g., Aroca,

2001; Ejdemo and Söderholm, 2011). The underlying idea is that positive spillovers – even in key sectors, such as manufacturing – are possible if forward or backward linkages are well-developed (Morris et al., 2012). Equally important for indirect employment opportunities would be the extractive region's ability to offer (and ideally produce) consumption goods and services that are demanded by mining workers (GIZ, 2016).

Secondly, it allows for the assessment of a greater source of concern: that related to the Dutch disease (Corden and Neary, 1982). While the exploitation of natural resources, namely of energy and mineral commodities, offers the possibility to increase a country's income, it could also lead to negative employment effects on other tradeable good sectors especially in the manufacturing industry – ultimately pushing the country toward deindustrialization.

Corden & Neary's Dutch disease model (1982) considers a booming tradeable sector, i.e., extractives, and the lagging tradeable sector i.e., agriculture and manufacturing, as well as the non-tradeable sector, i.e., including, services in general, retail and construction. The model operates mainly through two effects: The first is the *resource movement effect*: the booming sector increases its demand for production factors, including labor, drawing out labor from the lagging tradeable sector.

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<https://doi.org/10.1016/j.resourpol.2024.105136>

Received 5 September 2022; Received in revised form 8 February 2024; Accepted 29 May 2024

Available online 8 June 2024

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The direct impact of this effect is minimal as the extractive sector is a capital-intensive industry with low employment requirements. The second is the *spending effect*: The additional revenue brought about by the booming sector increases the demand for the non-tradeable sector, pushing prices up in this sector. This moves more labor away from the lagging sector toward the non-tradeable sectors. Moreover, the increased demand for the booming sector's commodities gives rise to a real exchange appreciation reducing the competitiveness of the lagging sectors' exports, further shrinking manufacturing industries, and moving production toward the non-tradeable and booming sectors. In short: The Dutch disease predicts an expansion of the extractive and service sectors at the expense of the agriculture and manufacturing industries.

As theorized by Krugman (1987), Matsuyama (1992), and Torvik (2001), the Dutch disease phenomenon could translate into a long-term adverse loss of competitiveness and ultimately growth because of the lost opportunities of learning-by-doing which take place predominantly in the manufacturing sector. Sachs and Warner (1995) were the first of several scholars to empirically establish the negative relation between natural resources and growth.¹ Yet in recent years, others have provided empirical evidence showing that long-term growth is not hampered by the abundance of natural resources² (Alexeev and Conrad, 2009; Brunnschweiler and Bulte, 2008; Cavalcanti et al., 2019; James, 2015; Lederman and Maloney, 2007). Furthermore, Gerelmaa & Kotani (2016) show that prior to the 1990s empirical evidence does support the Dutch disease hypothesis but not after. They explain that in the latter period (1990–2010), according to sector output, manufacturing had developed sufficiently— even in resource-rich countries – allowing countries to escape the effects of the Dutch disease.

The present study, thus, further investigates the effects of the Dutch disease but instead of focusing on sectoral output, it looks at employment growth in manufacturing and other non-mining tradeable sectors (i.e. refining industries and agriculture). Moreover, this is done at the country level – an important departure from previous studies e.g. Fleming and Measham (2014, 2015); Moritz et al. (2017); Weber (2012, 2014) which provide valuable insights into these issues from a strictly regional perspective. However, there are some limitations when analyses are done at this level. As explained in Fleming and Measham (2014), regional studies may not capture macroeconomic effects of mining expansion where the high exchange rate triggered by increasing mining exports could negatively impact manufacturing and other tradeable sectors across the country – as predicted by the Dutch disease. Likewise, the expansion of mining production within a mining region may generate additional demand for services and products *outside* the mining region which would eventually translate into job creation; a regional study, in this case, would also fail to capture positive spillovers which occur in the same country but outside the region analyzed.

While measuring negative and positive spillovers (if any) is a relevant consideration for policymakers, the main objective of this study is to explore whether the expansion of extractive activities, measured in mining employment changes, detrimentally affects non-mining tradeable sectors, i.e., manufacturing and agriculture, and promotes the growth of services, consistent with the Dutch disease expectations. In doing so, we distinguish the changes in mining employment driven by changes in domestic demand from those driven by external demand (exports). This differentiation is crucial as the adverse effects of mining expansion, as predicted by the Dutch disease, are tightly linked to foreign demand. Following Corden and

Neary (1982), this link exists not only because of the increased demand for production factors, including labor (the key focus of this study) predicted by the resource movement effect, but due to the mechanisms tied to external demand revenue (leading to the spending effect) and, more importantly, the rising exchange rates derived from the booming sector, which eventually hinder the competitiveness of other tradeable sectors. In contrast, the expansion of mining due to internal demand may not necessarily result in the same detrimental effects. While it could lead to some reallocation of production factors, including labor, the revenue generated by domestic demand would not be driven by high international prices. Therefore, the real exchange rate appreciation effects on other employment sectors would be absent. Overall, expected changes in employment demand stemming from domestic markets under the Dutch disease model of Corden and Neary are hypothesized not to be the same as when driven by external markets. Thus, exploring the different employment spillovers – triggered by foreign and domestic demand of mining output – may lead to a better understanding of the overall costs and opportunities of mining expansion at the national level.

Specifically, we explore the following questions:

- What are the effects of employment growth in the mining sector (due to changes in demand) on non-mining sectors' employment growth?
- Are the effects different if the source of demand is foreign, i.e., exports, instead of domestic?

To explore the effects that the expansion of the mining sector had on other non-mining sectors over the 2002–2014 period, we employ a sample of 40 countries from the WIOD dataset. We utilize the regression specification based on Moretti, extended in Fleming and Measham (2014). We further extend the analysis by adopting a GMM model approach to address potential endogeneity issues. While we control for aggregate changes in total employment in mining, we use two measures that capture employment changes in mining *only* due to demand: one that captures changes due to domestic demand and another that captures changes due to exogenous sources of demand, i.e. exports, following the method described in Foster Mc-Gregor (2019).

Finally, we test whether the adverse effects of Dutch disease are less likely to materialize in countries where the manufacturing sector output is high, as suggested in Gerelmaa and Kotani (2016). For this, we split the sample based on the level of manufacturing output and observe if the estimated elasticities differ across groups.

The paper is structured as follows: Section 2 discusses empirical methods and evidence surrounding employment spillovers of extractive industries, it is followed by Sections 3 and 4 in which the data and methodology are explained. Section 5 presents our empirical results and Section 6 concludes.

2. Empirical background: mining operations impacts on labor

To evaluate the potential of local economic impacts of mining projects, ex-ante assessments, i.e., projections, of indirect employment effects (also referred to as multipliers) are typically based on input-output projections. In this vein, the World Bank refers to a generic benchmark for multipliers: for every job in the mining sector (including oil and gas) between 1 and 4 additional jobs are created in the local economy³

¹ For a thorough survey of the literature of the Dutch disease and its evolution, see Badeeb et al. (2017).

² These studies employ different measures to capture abundance of natural resources, for example: primary sector exports over GDP and total exports (Lederman and Maloney, 2007); the GDP share of total natural resource and mineral resource exports (Brunnschweiler and Bulte, 2008) or oil and natural gas production as share of GDP (James, 2015). Furthermore, studies which have employed oil production or deposit specific measures (Alexeev and Conrad, 2009; Cavalcanti et al., 2011) have found overall positive effects on growth.

³ This figure is, however, based on South Australian mining operations (Eftimie et al., 2009). However, many such studies report local employment multipliers to be higher in less developed countries. For example, a GIZ meta-study reports that the median employment multiplier value of mining projects in Canada, Australia, Romania, New Zealand and South Africa is 1.92, but for Mali, Zambia and Tanzania the average reported value is 2.84 (GIZ, 2016). Another World Bank meta-study reports that indirect employment multipliers could be as high as 14 based on the Peruvian locality of Yanacocha (McMahon and Remy, 2002).

(Eftimie et al., 2009).

Scholarly works have also employed this method in region-specific cases.⁴ Aroca (2001) estimated that during the 1990s in the Chilean Region II (now Antofagasta), multipliers lay between 3.1 and 5.7 for private mining firms and between 1 and 1.7 for the rest of the mining sector. Ejdemo and Söderholm (2011) simulate the employment impacts of an iron ore project in Northern Sweden and find that for every 1 job in mining 1–1.5 jobs would be created elsewhere in the local economy. However, some consider that these projections are less reliable due to a number of assumptions that are necessary in input-output modeling, such as the unconstrained supply of labor and inputs – arguably an implausible scenario in most regions where extractive activities take place (let alone in a developing country). Weber (2012), for instance, points out that compared to ex-post econometric-based approaches, input-output projections in some regions in the US were upward biased. More importantly, input-output methods do not allow us to observe the crowd-out effects as contended by the Dutch disease.

For this, more recent scholarly works have focused on providing ex-post estimations of spillover effects of mining activities making use of different econometric techniques. Marchand (2012) analyzed mining operations in Western Canada during 1971–2006 and found that for every ten mining jobs created during boom periods, approximately three construction jobs, two retail jobs, and four and a half service jobs are created. Weber (2012, 2014) studied the effects of the greater natural gas production in the US in the 2000s concluding that gas production did little to crowd out manufacturing employment and that each mining-related job created more than one non-mining job. Another study in the Appalachia region of the US reached a similar conclusion: while there were marginal positive spillovers from the expansion of coal operations during the 1970s, there were no adverse effects on manufacturing (Black et al., 2005). Fleming and Measham (2014, 2015) analyzed local mining multipliers in different regions in Australia during 2001–2011. They find large positive impacts of mining expansion on employment in construction and services, i.e., professional, rental, and transport. They do not find crowding-out effects on manufacturing industries. Similar effects on services during the 2000s boom were identified in Northern Sweden by Moritz et al. (2017).

The above studies illustrate that at the regional level, indirect impacts on employment are concentrated in services; however, there are no signs of crowding out of manufacturing employment. This aligns with the findings of Cavalcanti et al. (2019) who compare oil-rich municipalities with municipalities without oil in Brazil from 1940 to 2000 in a quasi-experiment setup. Unlike previous studies, they do find that oil operations have a significant negative relation to agriculture employment; namely, they identify that workers in low-productivity agriculture migrated to formal service sectors, increasing the overall wealth of those municipalities (Cavalcanti et al., 2019).

Empirical studies at the regional level provide sound evidence that, at least in the context of industrialized and emerging countries, there are few reasons to believe that the Dutch disease would hamper manufacturing. Nonetheless, several dynamics at the national level cannot be captured at the regional level. As already mentioned, regional studies may not be able to capture aggregate Dutch disease effects, which therefore requires the effects to be studied at the national level.

⁴ Note that the terms ‘local’ and ‘regional’ refer to sub-national levels. In the studies here cited, ‘local’ refers to the immediate vicinity in which mining operations take place without having a formal geographical delimitation. Thus, ‘local’ makes reference to a small administrative unit, such as town, city or district – rather than a large area. Studies with region-specific focus refer to one or several specific administrative units, such as municipalities, states or provinces, where mining operations occur. For example, Ejdemo and Söderholm (2011) refer to the municipality of Pajala as ‘Northern Sweden’, whereas Marchand (2012) lists 8–17 provinces for the ‘Western Canada’ region.

3. Data

Our data consists of a 13-year panel and 40 countries from the WIOD database. The sample includes 36 developed economies and 7 developing economies. The sample also includes top countries in terms of energy and mineral commodities production: Australia, Brazil, Canada, China, Indonesia, India, Mexico, Norway, Russia, and the US. This, however, does not imply that the rest of the economies do not have a mining sector. For instance, mining operations in Finland, Sweden, Poland, and the Netherlands – included in the sample – have also important economic contributions during the period analyzed. All data employed in the following analysis are from WIOD unless indicated otherwise.⁵

4. Methodology

4.1. Mining employment growth measures

To be able to observe if employment expansion due to exports, as predicted by the Dutch disease, has different effects due to domestic demand, we follow the decomposition method of Foster-McGregor (2019) which in turn builds on Los et al. (2015), and Wang et al. (2017). The method allows us to observe changes in employment growth at the national level specifically due to changes in demand levels. The description in this sub-section is a synthesized version of the method in the study of Foster-McGregor (2019).

Based on a basic input-output setup, we begin by defining the reconstruction of vector $\mathbf{k}$ – a vector that contains employment requirements, i.e., the ratio of employment to gross output, for each sector:

$$\mathbf{k} = \hat{\mathbf{p}}\mathbf{L}^{-1}\mathbf{f} \quad (1)$$

where $\hat{\mathbf{p}}$ is a diagonalized matrix with the values of the $\mathbf{p}$ vector in the diagonal elements and zeros in all non-diagonal elements. The $\mathbf{p}$ vector expresses how much labor is required to produce an additional dollar in any given sector; this is derived by dividing all employment. A vector, $\mathbf{k}$, is then defined to capture the labor required in all stages of production of a final product. The matrix $\mathbf{L}$ is the well-known Leontief inverse matrix which shows by how much output rises in each sector of an economy when there is a one-dollar unit increase in final demand. $\mathbf{f}$ represents the final demand vector of any given sector. Thus, the reconstruction of the employment requirement $\mathbf{k}$ in (1) allows observing how employment labor requirements move as final demand changes, or as intermediate inputs changes according to the Leontief inverse matrix.

The decomposition then proceeds by constructing one plus the growth rate of employment, $\dot{\mathbf{k}}$, between two time periods (0 and 1), which is expressed as follows:

$$1 + \dot{\mathbf{k}} = \frac{\mathbf{k}_1}{\mathbf{k}_0} = \frac{\hat{\mathbf{p}}_1\mathbf{L}_1\mathbf{f}_1}{\hat{\mathbf{p}}_0\mathbf{L}_0\mathbf{f}_0} \quad (2)$$

To disentangle the drivers of employment growth in a given sector,

⁵ The World Input-Output Database (WIOD) offers annual time-series of world input-output tables spanning 43 countries and a model for the rest of the world from 2000 to 2014. Developed by a consortium led by Groningen University, the tables encompass 56 sectors and adhere to international classifications. The construction process involves linking national supply and use tables with international trade statistics, addressing challenges such as harmonization and import disaggregation. WIOD is a valuable resource for analyzing global value chains, regional specialization patterns, and job-task content changes (Timmer et al., 2015). However, a notable limitation in the context of our study is its sample coverage, which predominantly includes high- and middle-income economies.

the expression in (2) is further decomposed in this manner:

$$\frac{\hat{p}_1 L_1 f_1}{\hat{p}_0 L_0 f_0} = \frac{\hat{p}_1 L_1 f_1}{\hat{p}_0 L_1 f_1} \times \frac{\hat{p}_0 L_1 f_1}{\hat{p}_0 L_0 f_1} \times \frac{\hat{p}_0 L_0 f_1}{\hat{p}_0 L_0 f_0} \quad (3)$$

with the first term on the right-hand side in this expression showing effects due to employment requirements (i.e., the ratio of employment to gross output); the second, interindustry input requirements; and the last one, final demand.

Based on Foster-McGregor (2019),⁶ we then modify the decomposition to express the rates of employment growth:

$$\dot{k} \approx \ln\left(\frac{\hat{p}_1 L_1 f_1}{\hat{p}_0 L_1 f_1}\right) + \ln\left(\frac{\hat{p}_0 L_1 f_1}{\hat{p}_0 L_0 f_1}\right) + \ln\left(\frac{\hat{p}_0 L_0 f_1}{\hat{p}_0 L_0 f_0}\right) \quad (4)$$

We further decompose employment growth due to domestic production and exports as:

$$\dot{k}^D \approx \ln\left(\frac{\hat{p}_1 L_1 f_1^D}{\hat{p}_0 L_1 f_1^D}\right) + \ln\left(\frac{\hat{p}_0 L_1 f_1^D}{\hat{p}_0 L_0 f_1^D}\right) + \ln\left(\frac{\hat{p}_0 L_0 f_1^D}{\hat{p}_0 L_0 f_0^D}\right) \quad (5)$$

$$\dot{k}^E \approx \ln\left(\frac{\hat{p}_1 L_1 f_1^E}{\hat{p}_0 L_1 f_1^E}\right) + \ln\left(\frac{\hat{p}_0 L_1 f_1^E}{\hat{p}_0 L_0 f_1^E}\right) + \ln\left(\frac{\hat{p}_0 L_0 f_1^E}{\hat{p}_0 L_0 f_0^E}\right) \quad (6)$$

The measures we will use for the present analysis will then take the last term on the right-hand in (5) and (6) which express changes in the rate of employment growth due to changes in f^D , i.e., domestic final demand, and changes in f^E , i.e., final foreign demand or exports in the case of the mining sector. Since the latter terms, i.e., $\ln\left(\frac{\hat{p}_0 L_0 f_1^D}{\hat{p}_0 L_0 f_0^D}\right)$ and $\ln\left(\frac{\hat{p}_0 L_0 f_1^E}{\hat{p}_0 L_0 f_0^E}\right)$, only capture a subset of the changes in total mining employment, we also control for the effects of changes in total mining employment, i.e. $\dot{k}$ in (4), in some of our specifications.⁷

4.2. Econometric approach

The econometric approach we employ is based on Fleming and Measham (2014) – which is an extension of the elasticities model originally put forth by Moretti (2010) to quantify jobs in a city's tradable and non-tradable sectors due to exogenous increments in the number of jobs in the tradable sector.⁸ While this model was initially carried out at the city level, it has also been applied to analyze positive and negative spillover effects in studies at the regional level in mining regions in Australia, Canada, Sweden, and the US (Black et al., 2005; Fleming and

Measham, 2014, 2015; Marchand, 2012; Moritz et al., 2017).

The reduced form model in Moretti (2010) is:

$$\ln E_{c,t+1}^{NM} - \ln E_{c,t}^{NM} = \alpha + \beta \left(\ln E_{c,t+1}^M - \ln E_{c,t}^M \right) + \delta year_t + \varepsilon_{c,t} \quad (7)$$

where $\ln E_{c,t+1}^{NM} - \ln E_{c,t}^{NM}$ is the change in the growth rate of sector-specific (non-mining) employment, e.g., manufacturing or agriculture, in country c between time t and time $t + 1$. This would be equivalent to obtaining the yearly difference of employment growth of a given sector as captured in $\dot{k}$ as defined in equation (4). Similarly, $\ln E_{c,t+1}^M - \ln E_{c,t}^M$ is the change in the growth rate for mining employment. To control for structural changes (including macroeconomic adjustments or employment shocks) across years, the dummy variable $year$ is included. $\varepsilon_{c,t}$ is the error term.

Since we are interested in exploring the expansion of mining employment as the result of a booming export sector, we distinguish changes in the employment growth rate in the mining sector due to domestic demand $\ln E_{c,t+1}^{MDD} - \ln E_{c,t}^{MDD}$ from mining employment growth due to exports, $\ln E_{c,t+1}^{MX} - \ln E_{c,t}^{MX}$. These latter terms, thus, refer to the yearly difference of the last term on the right-hand side of equations (5) and (6), respectively.

To control for variables that may influence changes in employment indicators, we include a matrix of covariates, $\mathbf{X}$, which are country- and year-specific. This includes the GDP per capita growth which is incorporated to control for procyclical responses; naturally, some sectors will be more strongly correlated with economic growth, e.g., construction services, than others, e.g., public services. We also include the ratio of energy to non-energy commodities - which is calculated on a yearly basis as the ratio of fuel production values to nonfuel mining commodities production values.⁹ Furthermore, the ratio of energy to non-energy commodities is included to control for the heterogeneous impacts of energy commodities relative to non-energy commodities. As identified in Foster-McGregor et al. (2018), the former commodities have adverse economic effects and the latter do not. Specifically, the study highlights the long-term negative effects of energy commodity prices on per capita GDP, in contrast to non-energy commodity prices, which tend to have a positive long-term impact on per capita GDP. It would be plausible then that higher energy to non-commodity ratio is linked to lower employment in key sectors such as manufactures. This is grounded in the expectation that higher energy commodity ratios could influence economic conditions which in turn could have detrimental effects on employment in certain sectors. Therefore, including this ratio in our model allows us to account for potential variations in employment patterns related to energy and non-energy commodities.¹⁰

This results in the following:

$$\ln E_{c,t+1}^{NM} - \ln E_{c,t}^{NM} = \alpha + \zeta \left(\ln E_{c,t+1}^{MDD} - \ln E_{c,t}^{MDD} \right) + \gamma \left(\ln E_{c,t+1}^{MX} - \ln E_{c,t}^{MX} \right) + \beta \left(\ln E_{c,t+1}^M - \ln E_{c,t}^M \right) + \theta' \mathbf{X}_{c,t} + \delta year_t + \varepsilon_{c,t} \quad (8)$$

where ζ and γ are the parameters measuring the elasticity between mining employment due to domestic demand and due to exports and non-mining sector jobs. In other words, when jobs in the mining sector due to exports grow by 1%, the non-mining sector in the question sector grows by γ %.

Several papers following Moretti (2010) have empirically isolated the

⁶ For brevity purposes, the $\hat{p}$ matrix is not disaggregated as it is originally done in Foster-McGregor (2019).

⁷ Namely, we take $\dot{k} \approx \left(\frac{\dot{k}_1}{\dot{k}_0} \right)$ which approximates the rate of growth in total employment. By construction, this term is not collinear to the other terms in our estimation, i.e., $\ln\left(\frac{\hat{p}_0 L_0 f_1^D}{\hat{p}_0 L_0 f_0^D}\right)$ and $\ln\left(\frac{\hat{p}_0 L_0 f_1^E}{\hat{p}_0 L_0 f_0^E}\right)$, as these two derive from growth rates of different subsets of employment, $\dot{k}^D$ and $\dot{k}^E$, as expressed equations (5) and (6) respectively.

⁸ Several authors have pointed out that Moretti's (2010) method has several caveats, including the strong assumptions about fixed subnational employment shares on which the shift-share instrument relies, potentially resulting in inflated multipliers, especially in mining (Rajbhandari et al., 2022; Van Dijk, 2018). Yet, the current analysis circumvents these issues as instead of employing Moretti's (2010) shift-share instrument, we calculate Foster-McGregor's (2019) employment growth decomposition. Moreover, our estimates are considerably lower than those obtained by replicating Moretti's estimations, as done in Rajbhandari et al. (2022), suggesting our approach largely avoids the pitfalls concerning such a method.

⁹ GDP data is taken from the World Development Indicators. Fuel refers to coal, oil, and gas, whereas nonfuel mining commodities to aluminum, iron, steel, copper, gold, silver, nickel, tin, and zinc. Production values are expressed in thousands of USD. Data is from Thibault Fally's dataset on Commodity Trade which harmonizes and aggregates data from the British Geological Survey, and the Global Trade Analysis Project.

¹⁰ We employed other controls, such as regional dummies, in the specifications but were not significant so they were excluded from the estimation reported.

causal effect of the employment of the mining sector on employment in the non-mining sector by employing an instrumental variable approach. Fleming and Measham (2015), for instance, employ the number of coal seam wells as an instrument to better capture the change in mining employment due to the expansion of the mining projects and reduce the potential endogeneity issues. While we use decomposition measures that ‘isolate’ the effects of different demand sources on employment, these may still be subject to endogeneity. For this, we also address potential endogeneity in our analysis which may occur due to reverse causality as the direction of employment spillovers can occur bidirectionally. For instance, construction employment can expand as the result of the expansion of mining as building a mine naturally requires construction inputs. But it could also work in the opposite direction: the expansion of construction requires cement, steel, and other inputs which eventually result in higher demand for mining inputs and therefore mining employment. To address these concerns, we employ a system-GMM model, specifically the Arellano–Bover/Blundell–Bond dynamic panel model. The system-GMM model allows us to provide consistent estimators even in the presence of simultaneity and other issues such as unobserved heterogeneity in a country-level setup. This model incorporates lagged values of the dependent variable as internal instruments¹¹ (Roodman, 2009a). The inclusion of lagged variables and internal instruments enables us to address potential time lags in the responses of different sectors to changes in mining employment. This is a crucial consideration, as certain sectors may respond with a time delay to fluctuations in mining employment. The results are calculated using year-by-year changes, e.g., year-on-year employment or GDP growth rate, as indicated in the model. Our GMM estimations are also compared to estimations using OLS and Fixed Effects (FE) models.

For the last part of our analysis, where we test if the effects of Dutch disease are less likely to appear in countries with a higher manufacturing level, we split the sample based on the level of manufacturing output. A dummy variable is used to identify the countries with higher manufacturing output based on the initial median value of the share of manufacturing output in GDP (27%) using data from WIOD.¹² The estimation consists of a similar GMM approach to the one applied earlier; however, we interact this dummy variable with the explanatory variables of interest (i.e., changes in total mining employment, due to domestic demand, and due to exports). By including these interaction terms we can distinguish the differences in elasticities across the two groups while avoiding the issue of instrument proliferation we would have if we split the sample into two groups.¹³

¹¹ This approach aims to mitigate the impact of endogeneity and address the persistence of variables over time by employing internal instruments. These instruments are used to construct moment conditions that help eliminate bias in the parameter estimates (Roodman, 2009a). Namely, the system-GMM incorporates both moment conditions based on the levels of the variables and moment conditions based on differences or changes in the variables over time. This helps to address potential endogeneity issues and improve efficiency in estimating parameters in dynamic panel data models.

¹² Countries included in the manufacturing category (manufacturing = 1) are: Belgium, Brazil, China, Czech Republic, Germany, Spain, Finland, Hungary, Indonesia, India, Ireland, Italy, Japan, South Korea, Mexico, Romania, Slovakia, Slovenia, Sweden, and Turkey. Countries included in the non-manufacturing category (manufacturing = 0) are: Australia, Austria, Bulgaria, Canada, Switzerland, Cyprus, Denmark, Estonia, France, United Kingdom, Greece, Croatia, Lithuania, Latvia, Netherlands, Norway, Poland, Portugal, Russia, and the United States.

¹³ In a GMM estimation if the number of instruments is larger than the number of panels (in this case, countries), one incurs an issue of instrument proliferation. This results in the overfitting of endogenous variables reducing the reliability of the estimations (Roodman, 2009b). While one option is to restrict the number of instruments, GMM estimates are highly sensitive to how this is done. By estimating the GMM model with the whole sample using interaction terms, it is possible to distinguish effects across groups (as it would by splitting the sample) but with the same set of instrument restrictions, providing more efficient estimations.

5. Results

Table 1 presents summary statistics, offering insights into the variation within our sample, including growth rates of employment across different sectors. Notably, the standard deviation reveals the extent of variability in the *percentage change* in total employment across different sectors. For instance, Δ Manufacturing employment (log) illustrates in Table 1 the percentage change in total employment in total mining employment. Refining activities exhibit the highest variation, with a standard deviation of approximately 24%, followed by mining, agriculture, and construction, each displaying around 11%, 8.5%, and 8.4%, respectively. In contrast, the public services and services sectors show lower variation, with approximately 3.9% and 3.2%, respectively. This extends to growth rates in mining employment due to final domestic demand and exports, revealing substantial variation, as indicated by their standard deviations, which equate to approximately 21% and 18%, respectively. The range further underscores this variation. For instance, the growth rate of mining employment due to changes in domestic demand spans from -2.47 to 0.71 . The minimum value of -2.47 in logs implies a reduction of 91% each year, while the maximum value of 0.71 in logs signifies a growth rate of 71%. A similar, though less pronounced, range is observed for the growth rate of mining employment due to changes in exports, with a minimum of -1.03 and a maximum of 0.82 .

The average change in growth rates for all countries in each of the selected sectors can be seen in Fig. 1. The figure shows the contraction of mining employment due to foreign and domestic demand following the financial crisis and the expansion following recovery. This trend was closely followed by manufacturing, construction, and refining activities and to a lesser extent by services.

Fig. 1 also, shows that after commodity prices peaked in 2012, there was a contraction of mining employment due to domestic and foreign demand which was not seen in other sectors. Finally, while it is difficult to establish a clear average trend, the agriculture and public services sector experienced less movement and in the opposite direction after 2008.

Having discussed the overall trends, we will now present the results of the econometric analysis: first, the effects of aggregate mining employment on non-mining sectors employment followed by the effects of domestic demand and foreign demand separately.

Results in Table 2 indicate that the aggregate mining employment effects on manufacturing and refining employment are not significant in any of the models; except in the OLS model for refining employment in Column 6. In the case of mining employment expansion due to domestic demand, the results for the OLS and FE models (Columns 1 and 2) indicate a positive relationship with the creation of manufacturing jobs, significant at the 1% and 10% level, respectively. Namely, the results indicate that a 10% increment in mining employment due to domestic demand correlates with an additional 0.27–0.31% employment in the manufacturing sector.¹⁴ This effect, however, loses significance in the GMM model (Columns 3, 4 and 5) – a somewhat expected result due to potential simultaneity bias which is not addressed in the OLS and FE models.

More importantly, however, is that our results do not show any significant negative effects on mining employment growth due to foreign demand. Controls indicate that GDP is positively linked to manufacturing employment – a result that is consistently positive in all models (Columns 1–5). Lastly, we do not find a negative effect on the

¹⁴ While our preferred specification is the GMM, it is worth mentioning about the interpretation of the OLS/FE results: Given the low levels in mining employment, a 10% change in absolute number of jobs is relatively small, but a 0.27% increase in manufacturing jobs would be much larger as well as its overall impacts as manufacturing is more important in the economy. In this case, however, due to potential simultaneity, these elasticities are best interpreted as a correlation.

Table 1

Summary statistics.

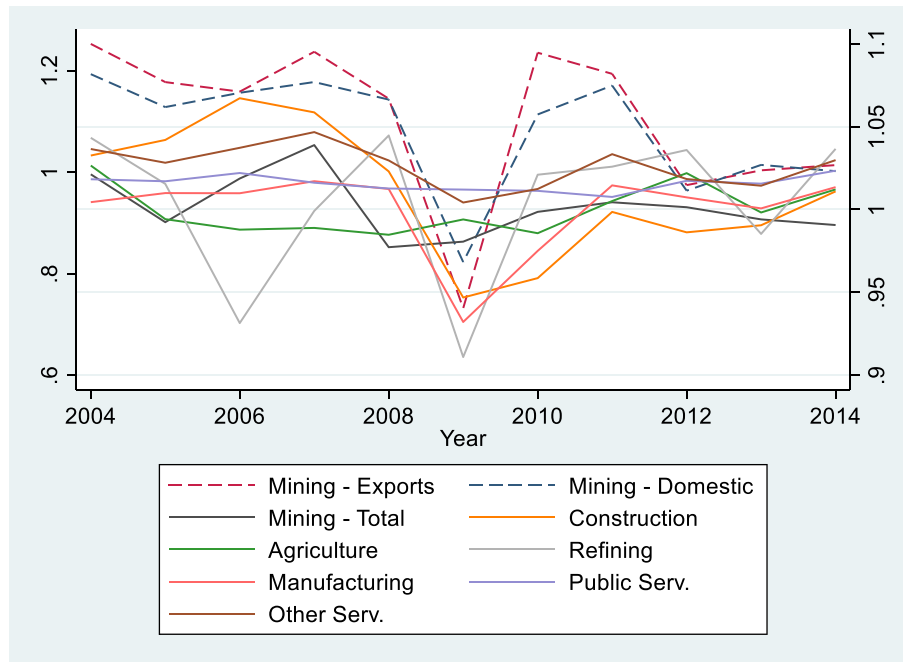
Variable	N	Mean	SD	Min	Max
Δ Manufacturing employment (log) ^a	480	−0.004	0.109	−0.692	0.418
Δ Mining employment due to changes in domestic demand (log) ^b	480	0.071	0.209	−2.467	0.710
Δ Mining employment due to changes in exports (log) ^c	480	0.092	0.179	−1.033	0.817
Δ Manufacturing employment (log) ^a	480	−0.008	0.051	−0.268	0.158
Δ Refining activities employment (log) ^a	474	−0.014	0.236	−1.689	1.179
Δ Agriculture employment (log) ^a	480	−0.010	0.085	−0.362	0.760
Δ Construction employment (log) ^a	480	0.008	0.084	−0.414	0.338
Δ Services employment (log) ^a	480	0.019	0.039	−0.134	0.316
Δ Public services employment (log) ^a	480	0.015	0.032	−0.087	0.162
GDP per capita (log)	480	9.971	0.932	6.827	11.425
Energy to non-energy mining production value ratio (log)	480	−0.807	2.881	−4.605	15.500
Share of manufacturing output in total gross output	480	0.255	0.081	0.080	0.490
Observations	480				

Notes.

^a Refers to the logarithm of the ratio of total employment in a given sector in year 1 (k_1) to total employment in year 0 (k_0): $k \approx \ln\left(\frac{k_1}{k_0}\right)$.

^b Refers to logarithm of the third term on the right-hand side of formula (5): $\ln\left(\frac{\hat{p}_0 L_0 f_1^D}{\hat{p}_0 L_0 f_0^D}\right)$.

^c Refers to logarithm of the third term on the right-hand side of formula (6): $\ln\left(\frac{\hat{p}_0 L_0 f_1^E}{\hat{p}_0 L_0 f_0^E}\right)$.

**Fig. 1.** Total employment (in logs) of mining and selected sectors, and the subset of employment (in logs) due to domestic demand and exports.

Note: The y-axis on the left corresponds to the scale of total employment in mining (total), construction, agriculture, refining activities, manufacturing, public services, and services; and the y-axis on the right to the subset of employment in the mining sector either due to exports or to domestic demand.

Source: Author's elaboration with WIOD data.

ratio of energy to non-energy commodities on manufacturing employment; the coefficient is positive but is only significant in the FE estimation (Column 2).

Based on our results, thus, we find no evidence to support the notion that expanding domestic employment in the mining sector would lead to reduced employment in the manufacturing sector, as hypothesized in the Dutch disease. Our results moreover, align with the body of regional studies that do not find any detrimental effects on manufacturing activities (e.g. Black et al., 2005; Cavalcanti et al., 2019; Moritz et al., 2017).

Regarding refining activities, coefficients in Columns 5–8 from Table 2 show that, on average, growth of employment in mining – regardless of the source of demand or as an aggregate – does not translate

into significant negative or positive effects for that sector's employment. Likewise, the energy-to-non-energy ratio is insignificant. Lastly, GDP per capita growth had a negative effect significant at the 10% in the FE model (column 6) – the effect however disappears in the GMM models in Columns 8 and 9 and becomes significant in the last GMM where all controls are introduced.

The above suggests that mining activities are decoupled from mineral processing activities – which is not necessarily negative per se: in primary sectors, including mining, processing activities do not always generate greater domestic value-added (Korinek, 2020). However, from a policy perspective, it should not be assumed that the expansion of mining projects will lead to higher employment opportunities in

Table 2
Results for manufacturing and refining activities.

	1	2	3	4	5	6	7	8	9	10
	Δ Manufacturing employment (log)					Δ Refining activities employment (log)				
	Pooled OLS	Fixed Effects	GMM	GMM		Pooled OLS	Fixed Effects	GMM	GMM	
Δ Mining employment – total (log)	0.031 –1.63	0.033 –1.26		0.007 –0.26	0.0076 –0.27	0.322 ^a –1.86	0.314 –1.46		0.298 –1.11	0.240 –0.85
Δ Mining employment - Domestic demand (log)	0.031 ^c –2.85	0.027 ^a –1.92	0.022 –0.91	0.020 –0.83	0.020 –0.83	0.021 –0.39	0.026 –0.54	–0.241 –0.98	–0.098 –0.64	–0.168 –0.73
Δ Mining employment - Exports (log)	0.014 –0.94	0.011 –0.82	0.003 –0.21	0.004 –0.26	0.004 –0.27	–0.005 –0.06	0.003 –0.03	–0.134 –1.07	–0.035 –0.29	–0.077 –0.62
Δ GDP p.c. (log)	0.333 ^c –3.52	0.572 ^c –7.22	0.800 ^c –9.01	0.892 ^c –4.38	0.874 ^c –4.4	–0.8 –1.55	–1.613 ^a –1.72	3.55 –1.19	0.298 –0.2	4.274 ^a –1.91
Energy to non-energy ratio (log)	0.001 –0.89	0.005 ^c –10.65			0.001 –0.54	–0.003 –0.37	–0.008 –0.82			0.001 –0.89
Δ Manufacturing employment (log), t-1	–0.435 ^c	–0.454 ^c	–0.351 ^c	–0.359 ^c	–0.360 ^c					
Δ Refining activities employment (log), t-1	–6.77 ^c	–11.03 ^c	–5.42 ^c	–5.73 ^c	–5.86 ^c	–0.38 ^c –4.96	–0.39 ^c –9.43	–0.135 –1.29	–0.186 ^a –1.80	–0.223 ^b –2.10
Constant	–0.053 ^c –4.67	0.005 –0.71	0.005 –0.52	0.005 –0.48	0.005 –0.53	0.036 –0.79	0.057 –1.68	–0.146 ^a –1.78	0.0003 0	–0.163 ^b –2.14
N	440	440	440	440	440	427	427	427	427	427
R-sq	0.472	0.506				0.222	0.227			
adj. R-sq	0.452	0.487				0.192	0.197			
Time effects	Yes	Yes				Yes	Yes			
Fixed effects	No	Yes				No	Yes			
Sargan p-value			0.000	0.000	0.000			0.020	0.003	0.008
Hansen p-value			0.313	0.351	0.273			0.476	0.685	0.693
AR1 p-value			0.005	0.004	0.004			0.021	0.006	0.023
AR2 p-value			0.769	0.832	0.854			0.804	0.402	0.053
Instruments (j)			30	36	37			30	36	37

t-statistics in parentheses.

^a $p < 0.10$.

^b $p < 0.05$.

^c $p < 0.01$.

downstream sectors, as our results demonstrate.

Table 3 indicates that changes in total mining employment do not have any significant effects on employment within agriculture (including forestry and fisheries). The same holds for changes in mining employment due to domestic demand where no significant effects are found possibly due to the lack of strong linkages and/or complementarities among natural resource sectors (i.e., mining and agriculture).

However, when it comes to changes in mining employment due to exports, the results suggest a negative impact on agriculture jobs. Coefficient in Columns 1 to 5 are significant at the 5% and 10% level depending on the model. Based on the GMM models (Columns 3 to 5) the negative elasticities suggest that a 10% increment in mining employment due to exports is linked to a reduction of agricultural employment equivalent to 0.7%. The magnitude however will depend on the size of the economy. To illustrate this, consider Australia and Russia in 2014, where employment in agriculture amounted to 297 thousand and 11,828 thousand workers, respectively. If the mining sector were to expand by 10%, our model predicts a reduction of approximately 2.08 and 82.79 thousand workers in agriculture for Australia and Russia, respectively.

Our findings align with previous empirical studies which suggest that the expansion of mining activity in a region can crowd out employment in the agricultural sector (Fleming and Measham, 2015). Yet, the coefficients in our estimations indicate that at the national level, effects are considerably smaller if compared to regional effects.¹⁵

GDP per capita growth coefficients – insignificant in all models –

¹⁵ Using two-stage least square estimations, Fleming and Measham (2015), calculate a –0.34 elasticity between these sectors in selected regions in Australia; in other words, a 10% expansion of coal seam gas employment would lead to a reduction of 3.4% reduction of employment in the agricultural sector.

show that the sector has little procyclicality – an expected result given the inelasticity in demand for agricultural products.

Moving forward to construction, our results indicate that changes in total mining employment have no significant impact on employment in this sector, as seen in Columns 6 to 7, and 9 to 10 in Table 3. The same is observed for changes due to exports (Columns 6–10).

Nevertheless, changes in mining employment due to domestic demand show a positive and significant effect on construction jobs across all models; generally speaking, housing demand and infrastructure investments are highly linked to mining activity (Fleming and Measham, 2014). According to the elasticities in Table 3, a 10% increment in mining employment due to domestic demand are linked to an increment of 0.5%–0.6% in construction employment based on the OLS and FE estimations (significant at the 1% and 10% level, respectively). More importantly, our preferred specification (GMM) suggests that a 10% increment in mining employment due to domestic demand is linked to an increment of 1.1%–1.3% in construction employment (significant at the 5% and 1% level, depending on the model). To provide context, considering Canada and Mexico in 2014, where construction employment figures were 1432.82 thousand and 5560.61 thousand workers, respectively, the model predicts an increase of approximately 15.76–61.16 thousand workers in construction employment if the mining sector due to domestic demand were to expand by 10%. While the effects are consistent with those identified in regional studies, they are considerably smaller.¹⁶

Lastly, while the coefficients on GDP per capita growth exhibit positive and significant associations across the OLS and FE models,

¹⁶ In Black et al. (2005), the calculated elasticity for construction is around 0.5 for two boom periods (1970s and 1980s) in the US.

Table 3
Results for agriculture and construction activities.

	1	2	3	4	5	6	7	8	9	10
	Δ Agriculture employment (log)					Δ Construction activities employment (log)				
	Pooled OLS	Fixed Effects	GMM	GMM	GMM	Pooled OLS	Fixed Effects	GMM	GMM	GMM
Δ Mining employment – total (log)	0.008 –0.18	0.006 –0.11		–0.050 –0.66	–0.049 –0.66	0.030 –1.21	0.037 –1.52		–0.080 –0.93	–0.081 –0.91
Δ Mining employment - Domestic demand (log)	0.004 –0.35	0.003 –0.33	–0.166 –0.98	–0.071 –1.27	–0.072 –1.28	0.061 ^c –3.75	0.052 ^b –2.42	0.128 ^c –3.15	0.114 ^b –2.66	0.116 ^b –2.7
Δ Mining employment - Exports (log)	–0.0436 <i>a</i>	–0.0440 <i>b</i>	–0.0711 <i>a</i>	–0.0704 <i>b</i>	–0.0703 <i>b</i>	0.009	0.003	0.020	0.000	0.001
Δ GDP p.c. (log)	–1.7 0.0611 –0.33	–2.35 0.127 –0.45	–1.73 –0.605 –0.68	–2.41 0.288 –0.91	–2.4 0.285 –0.9	–0.56 0.725 ^c –5.13	–0.17 1.202 ^c –5.39	–0.72 0.544 –1.56	–0.01 0.507 –1.63	–0.02 0.479 –1.52
Energy to non-energy ratio (log)	–0.001 –0.4	–0.003 –1.08			–0.00116 ^a –1.83	0.002 –0.88	0.011 ^c –12.17			0.002 –1.01
Δ Agriculture employment (log), t-1	–0.477 <i>c</i>	–0.478 <i>c</i>	–0.560 <i>c</i>	–0.480 <i>c</i>	–0.479 <i>c</i>					
Δ Construction activities employment (log), t-1	–3.90	–4.53	–3.30	–5.62	–5.56	–0.233 <i>c</i>	–0.290 <i>c</i>	–0.0108	–0.151	–0.155
Constant	–0.0111 –0.56	0.00000217 0	0.108 –1.62	0.0552 –1.67	0.0541 –1.66	–0.0285 ^a –1.77	0.00692 –0.75	–0.0529 ^a –1.84	–0.0467 ^a –1.98	–0.0457 ^a –1.92
N	440	440	440	440	440	440	440	440	440	440
R-sq	0.263	0.264				0.327	0.396			
adj. R-sq	0.235	0.237				0.302	0.374			
Time effects	Yes	Yes				Yes	Yes			
Fixed effects	No	Yes				No	Yes			
Sargan p-value			0.375	0.044	0.042			0.058	0.007	0.008
Hansen p-value			0.348	0.363	0.357			0.421	0.387	0.379
AR1 p-value			0.024	0.002	0.002			0.035	0.059	0.064
AR2 p-value			0.536	0.648	0.668			0.866	0.725	0.718
Instruments (j)			30	36	37			30	36	37

t-statistics in parentheses.

^a $p < 0.10$.

^b $p < 0.05$.

^c $p < 0.01$.

indicating a procyclical behavior of the sector as anticipated, they do not demonstrate significance in our preferred specification.

Table 4 presents the results for services, which encompasses two aggregate sectors – services (which includes retail and various professional services) and public services. First, we find that for changes in total mining employment, there are no significant effects on services and public services in the preferred GMM specifications (Columns 3–5 and 8–10, respectively). Similarly, the coefficients for mining employment changes due to domestic demand are insignificant across specifications in both services and public services.

In the case of services, changes in mining employment due to exports exhibit a positive and significant elasticity in the OLS, FE, and GMM models in Columns 1 to 4. However, this is not the case for the GMM model in Column 5, where the introduction of the energy to non-energy ratio, a non-significant predictor of our dependent variable, renders the result non-significant. Referring to the results in Table 4, a 10% increase in mining employment due to domestic demand corresponds to a 0.3% increase in services employment based on the OLS and FE estimations (Columns 1 and 2), significant at the 5% and 1% levels, respectively. GMM models (Columns 3 and 4) also indicate a similar effect, with a 10% rise in mining employment due to exports resulting in a 0.2% increase in service employment, significant at the 5% and 10% levels, respectively. To illustrate the magnitude, consider Brazil and Indonesia in 2014, where employment in services amounted to 29,056.33 thousand and 44,079.65 thousand workers, respectively. If the mining sector were to expand by 10%, our model predicts an increase of approximately 58.11 and 88.16 thousand workers in services for Brazil and Indonesia, respectively.

Our findings align with previous empirical studies which find positive effects for professional and retail services (e.g. Marchand, 2012; Moritz et al., 2017). Moreover, this positive effect on service

employment provides support to the predicted effects of Dutch disease (i.e., expansion of services as a result of higher mining exports), but the effects are not particularly pronounced at the national level.

Moving on to public services, the results are different from those for construction and services. Although the OLS and FE models indicate a positive and significant elasticity (at 1% and 5% respectively), elasticities in the GMM models are not significant. This result aligns with previous studies that have failed to find any mining employment spillovers on government employment (Moritz et al., 2017).

Lastly, the ratio of energy to non-energy commodities is not significant in most models – including the GMM specifications. While GMM models without controls (Columns 3 and 8) do not show significance, the rest of results suggest procyclicality in employment for services and public services. This is indicated by the positive and significant coefficients for GDP per capita growth in the OLS and FE (Columns 1–2 and 6–7) as well as the GMM models (Columns 4–5 and 9–10), underscoring the sensitivity of employment in these sectors to economic growth.

5.1. Differences across countries according to manufacturing levels

In this last part of the analysis, we explore differences in elasticities across countries with a higher level of manufacturing. The dummy variable *manufacturing* is assigned to the group with a higher level of manufacturing output.¹⁷ To adequately interpret results in this sub-

¹⁷ Recall that this is done on the basis of median values of manufacturing output in 2000. This means that those countries above the median value of manufacturing have the value of 1, and labeled *manufacturing*, otherwise, they have the value of 0.

Table 4

Results for services: services, and public services.

	1	2	3	4	5	6	7	8	9	10
	Δ Services employment (log)					Δ Public services activities employment (log)				
	Pooled OLS	Fixed Effects	GMM	GMM	GMM	Pooled OLS	Fixed Effects	GMM	GMM	GMM
Δ Mining employment – total (log)	0.0372 ^b –2.22	0.0388 ^a –2.02		0.015 –0.57	–0.045 –0.99	0.0172 –1.33	0.0172 ^a –1.82		0.012 –0.27	0.012 –0.27
Δ Mining employment - Domestic demand (log)	0.005 –0.60	0.002 –0.24	–0.012 –0.34	–0.012 –0.82	–0.005 –0.19	0.001 –0.13	0.000 –0.08	–0.032 –0.60	–0.039 –0.77	–0.039 –0.77
Δ Mining employment - Exports (log)	0.030 ^c –2.75	0.027 ^c –2.75	0.019 ^a –1.68	0.025 ^b –2.19	0.009 –0.71	0.028 ^c –2.64	0.028 ^b –2.51	0.020 –1.02	0.020 –1.12	0.020 –1.11
Δ GDP p.c. (log)	0.210 ^b –2.36	0.445 ^c –3.49	0.68 –1.61	0.559 ^c –3.11	0.713 ^b –2.60	0.166 ^b –2.56	0.209 ^a –1.99	0.387 –0.74	0.478 ^a –1.75	0.479 ^a –1.72
Energy to non-energy ratio (log)	0.001	0.00274 ^c			0.000	0.000	0.001			0.000
Δ Services employment (log), t-1	–1.05 –0.432 ^c	–5.47 –0.448 ^c		–0.386 –0.323 ^c	–0.60 –0.455 ^c	–0.94	–1.34			–0.55
Δ Public services activities employment (log), t-1	–7.43	–14.7	–3.27 ^c	–5.2 ^c	–3.54 ^c	–0.426 ^c	–0.437 ^c	–0.219	–0.302	–0.306
Constant	–0.00885	0.00552	–0.0149 ^c	–0.0186 ^b	–0.0244 ^b	–0.0187 ^b	0.00319	–0.00926	–0.00914	–0.00868
N	440	440	440	440	440	440	440	440	440	440
R-sq	0.356	0.401				0.242	0.253			
adj. R-sq	0.332	0.378				0.213	0.225			
Time effects	Yes	Yes				Yes	Yes			
Fixed effects	No	Yes				No	Yes			
Sargan p-value			0.075	0.027	0.122			0.133	0.098	0.094
Hansen p-value			0.304	0.607	0.327			0.354	0.511	0.509
AR1 p-value			0.022	0.000	0.017			0.015	0.006	0.006
AR2 p-value			0.728	0.870	0.537			0.811	0.570	0.560
Instruments (j)			30	36	37			30	36	37

t-statistics in parentheses.

^a $p < 0.10$.^b $p < 0.05$.^c $p < 0.01$.

section consider the following: the GMM models presented contain non-interaction terms of our variables of interest, i.e., changes in mining employment - either total or due to changes in domestic demand or exports. Coefficients of non-interacted terms, e.g., Δ Mining employment – total (log), have a direct interpretation; they are the estimated elasticity for the group of countries with relatively lower manufacturing output. The interaction terms with the manufacturing dummy, e.g. Δ Mining employment – total (log) * manufacturing, between countries with lower and higher levels of manufacturing. The elasticity in this latter group, then, is derived from the sum of the estimated coefficient of the non-interaction term (i.e. the estimate for the countries with lower manufacturing) and the estimated coefficient of the interaction term.

Having explained the interpretation of interaction terms, we proceed to discuss the results in Tables 5–7. For employment in manufacturing and refining activities, we do not find any statistical difference among the two groups as can be seen in Columns 1–6 in Table 5. This reaffirms our earlier findings and aligns with existing empirical evidence, which also fails to identify evidence of negative spillovers in manufacturing as mining employment expands.

These results underscore the importance of not assuming that the expansion of mining employment due to higher demand will necessarily lead to positive spillovers in downstream industries, as previously mentioned.

Our results suggest that mining employment expansion does crowd out employment in agriculture but only in countries with a lower level of manufacturing – as seen in Table 6. The elasticity of mining employment due to exports becomes significant at the 5% and 10% level. In Column 3, the total mining employment coefficient remains negative but is not significant. The results for the interacted terms remain insignificant in all cases, implying that the crowding-out effect does not appear in countries with higher manufacturing output.

According to Column 3 in Table 6, a 10% increment in mining employment due to exports would be linked to a 7% reduction in agriculture employment in countries with lower shares of manufacturing – which is the same effect calculated in the estimation with all countries (Table 3). Fleming and Measham (2015) find similar results in Queensland, Australia for the 2001–2011 period, and explain that workers in agriculture moved to the local mining sector, services, or construction. While it is not possible to assume this happened across our sub-sample of countries, it could be a plausible explanation. The exact mechanisms behind these labor movements happening in countries with lower shares of manufacturing are not entirely clear. Yet, a possibility would be that in more manufacturing-intensive countries there already has been an absorption of agricultural workers in the manufacturing sector, and there is fewer room for this type of dynamic. Overall, these findings support the notion that countries with a stronger manufacturing

Table 5

Results – Manufacturing and refining activities employment with interactions term for countries with higher manufacturing.

	1	2	3	4	5	6
	Δ Manufacturing employment (log)			Δ Refining activities employment (log)		
Δ Mining employment – total (log)			–0.019			–0.321
Δ Mining employment – total (log) * <i>manufacturing</i>			–0.14			–0.6
			0.022			0.299
			–0.10			–0.47
Δ Mining employment - Domestic demand (log)	0.017	0.023	–0.035	–0.007	0.030	0.131
	–0.52	–0.55	–0.76	–0.06	–0.27	–1.08
Δ Mining employment - Domestic demand (log) * <i>manufacturing</i>	–0.052	–0.038	–0.072	0.265	0.213	0.367
	–1.1	–1.16	–1.1	–0.67	–0.72	–0.87
Δ Mining employment - Exports (log)	–0.019	0.007	–0.030	–0.007	–0.050	–0.022
	–0.84	–0.32	–0.9	–0.03	–0.23	–0.11
Δ Mining employment - Exports (log) * <i>manufacturing</i>	0.044	–0.014	0.032	–0.101	–0.043	–0.262
	–0.96	–0.35	–0.8	–0.28	–0.08	–1.25
Energy to non-energy value ratio (log)		0.017	0.001		0.056	0.0991 ^b
		–1.14	–0.1		–1.49	–2.3
Energy to non-energy value ratio (log) * <i>manufacturing</i>		–0.010	0.000		–0.063	–0.112
		–0.66	–0.03		–1.1	–1.54
Δ GDP p.c. (log)	0.441	0.208	1.152 ^c	–0.094	1.956	5.445
	–1.21	–0.48	–2.84	–0.04	–0.55	–1.2
Δ Manufacturing employment (log), t-1	–0.202 ^b	–0.066	–0.485 ^b			
	–2.38	–0.31	–2.29			
Δ Refining activities employment (log), t-1				0.233 ^b	0.043	–0.157
				–2.03	–0.19	–1.02
Constant	0.02	0.042 ^a	–0.012	0.00704	–0.0531	–0.172
	–0.89	–1.67	–0.8	–0.07	–0.47	–0.84
Observations	440	440	440	427	427	427
Sargan p-value	0.004	0.036	0.359	0.727	0.410	0.479
Hansen p-value	0.390	0.580	0.624	0.411	0.368	0.720
AR1 p-value	0.003	0.018	0.024	0.013	0.093	0.049
AR2 p-value	0.062	0.169	0.921	0.281	0.614	0.975
Instruments (j)	37	36	37	37	36	37

t-statistics in parentheses.

^a p < 0.10.^b p < 0.05.^c p < 0.01.

sector are less likely to experience effects linked to Dutch disease.

In line with the above, results concerning the construction sector also show that effects are significantly more likely in countries with lower levels of manufacturing. The expansion of construction services in response to mining employment due to domestic demand appears strong in Columns 4–6 in Table 6. Akin to agriculture, elasticities appear positive and significant only in the non-interacted terms. This would imply that for a 10% increment in mining employment due to domestic demand, there would be an increment in construction employment between 1.1% and 1.4%. The coefficients for the group where manufacturing is higher are insignificant and have a net effect close to zero. This underscores that the expansion of services, including construction, is less likely to appear in the group with higher manufacturing output.

Turning to services, we observe that changes in total mining employment did not correspond to an increase in service jobs in either subgroup. Yet, elasticities in mining employment due to exports showed significance within the lower-manufacturing group, i.e., the non-interacted coefficient for changes in mining employment due to exports is significant across Columns 2 and 3 in Table 7 at the 1% and 10% level, respectively. This suggests that a 10% rise in mining employment due to exports would lead to a 0.4% expansion in service employment—a stronger effect compared to the estimations for all countries in Table 4. However, it is important to note that the GMM model without controls in Column 1 of Table 7 is not significant, indicating less robustness and magnitude compared to those observed for construction. This underscores the distinction in dynamics driven by domestic and external demand when analyzing changes in mining employment.

To summarize, our results concerning agriculture and services confirm that Dutch disease-related effects are less likely to materialize in countries with a higher level of manufacturing activity as suggested by

Gerelmaa and Kotani (2016). However, these effects do not involve manufacturing, which is the main concern associated with the Dutch disease.¹⁸ It should be noted that because countries in our sample are mainly developed and emerging economies, industrial development is probably strong enough to withstand negative externalities linked to higher demand for mining output. Therefore, we do not find any statistical effects in any of the groups – even if the countries in our sample are below the established manufacturing output threshold used to create the two groups. The results should not be generalized to economies with very underdeveloped manufacturing sectors; it is reasonable to expect different outcomes when using a sample that includes countries with very low levels of manufacturing.

6. Conclusion

The objective of this paper is to shed light on the positive and negative impacts of expanding mining activity by focusing on

¹⁸ We conducted additional tests, including running our models with only Δ Mining employment – total (log) (excluding the subcomponents for employment due to domestic demand and exports), but it did not show significance in any of the GMM models – our preferred specification. Incorporating additional control variables, such as dummies for developing countries and extractive concentration in exports, did not yield significant results. Further tests involving the share of value in manufacturing and agriculture showed insignificant effects, except for the share of manufacturing value in the total economy, which exhibited expected significance at the 90% level in one of the GMM models explaining differences in services. The results from this particular model revealed the disappearance of the previously observed significant positive effect on service employment, consistent with our findings in Table 7, where manufacturing effects are modeled as a dummy variable.

Table 6

Results – Agriculture and construction employment with interaction terms for countries with higher manufacturing.

	1	2	3	4	5	6
	Δ Agriculture employment (log)			Δ Construction employment (log)		
	GMM	GMM	GMM	GMM	GMM	GMM
Δ Mining employment – total (log)			–0.109			0.003
			–0.77			–0.04
Δ Mining employment – total (log) * <i>manufacturing</i>			0.092			–0.120
			–0.41			–0.77
Δ Mining employment - Domestic demand (log)	–0.026	–0.012	0.000	0.112 ^c	0.124 ^b	0.145 ^c
	–0.93	–0.14	0	–2.77	–2.6	–2.75
Δ Mining employment - Domestic demand (log) * <i>manufacturing</i>	–0.087	–0.054	–0.082	–0.056	0.054	0.030
	–0.69	–0.59	–0.87	–0.87	–0.54	–0.28
Δ Mining employment - Exports (log)	–0.0659 ^a	–0.0618 ^b	–0.0676 ^b	0.006	0.041	0.026
	–1.77	–2.11	–2.12	–0.15	–0.8	–0.52
Δ Mining employment - Exports (log) * <i>manufacturing</i>	–0.003	–0.013	–0.036	0.012	–0.059	–0.023
	–0.03	–0.15	–0.68	–0.2	–0.76	–0.39
Energy to non-energy mining value ratio		–0.015	0.011		0.021	0.002
		–1.08	–0.81		–1.54	–0.16
Energy to non-energy mining value ratio * <i>manufacturing</i>		–0.005	–0.018		–0.007	0.007
		–0.26	–1.55		–0.37	–0.52
Δ GDP p.c. (log)	0.418	0.760	0.809	0.592	1.516 ^b	1.875 ^b
	–0.48	–0.65	–0.65	–1.3	–2.06	–2.62
Δ Agricultural employment (log), t-1	–0.435 ^c	–0.322 ^b	–0.479 ^b			
	–3.85	–2.38	–2.45			
Δ Construction employment (log), t-1				0.0242	0.152	0.104
				–0.15	–0.78	–0.38
Constant	0.0467	0.00511	0.0401	–0.0327	–0.0473	–0.049
	–1.57	–0.1	–0.87	–1.41	–1.14	–1.4
Observations	440	440	440	440	440	440
Sargan p-value	0.121	0.110	0.089	0.015	0.098	0.284
Hansen p-value	0.658	0.589	0.691	0.536	0.315	0.108
AR1 p-value	0.003	0.006	0.044	0.029	0.004	0.024
AR2 p-value	0.766	0.264	0.827	0.781	0.346	0.543
Instruments (j)	37	36	37	37	36	37

t-statistics in parentheses.

^a $p < 0.10$.^b $p < 0.05$.^c $p < 0.01$.

employment changes linked to foreign and domestic demand at the national level since most of the evidence has focused primarily on regional effects.

We do not find any evidence to support the notion that the manufacturing sector is hampered by the expansion of mining employment – the main concern linked to the Dutch disease. These results hold even across countries with relatively low manufacturing output, such as Australia, Canada, Netherlands, or Russia. Similarly, we do not find any negative or positive effects on employment linked to downstream industries, such as refining activities.

However, it is not possible to rule out the adverse effects of the Dutch disease in manufacturing in mineral-rich countries with an incipient level of industrial development, such as Ghana or Peru. In this regard, it should be considered that our sample mainly covers high-income countries, and some emerging economies, such as Brazil, Russia, Turkey, and Mexico. Therefore, our findings should not be extended to countries with limited industrial capacity.

Although the biggest concern for many policymakers is the loss of employment in the manufacturing sector, we do find negative spillovers in other tradeable sectors, i.e., agriculture (which includes forestry, and fisheries). Yet, this strong effect appears only in countries with a relatively low level of manufacturing output. Based on the Australian case, Fleming and Measham (2015) hypothesize that this is the result of agricultural workers migrating to other service sectors (or even mining) but could also be attributed to investments in labor-saving technologies in agriculture. They highlight, though, the need for further research in that regard as the underlying mechanisms are not yet clear. Likewise, it

also remains unclear how this crowd-out effect would play out in less developed countries. Indeed, there is great heterogeneity in the added value of primary sectors, land use, and agricultural productivity. Therefore, the extent to which negative spillovers could hamper the economy would depend on how strategic and well-developed the natural resource sectors are in each country.

In terms of positive spillovers associated with the expansion of mining employment, our findings suggest a consistent and important positive effect on the construction sector. It is worth mentioning that the effect was linked to mining employment changes due to domestic demand and not exports. These findings are supportive of previous empirical studies where it is found that construction is strongly linked to mining activity in selected regions in Australia or the US. However, under closer inspection, the effects, again, are seen only in the group with a relatively lower level of manufacturing output.

Furthermore, the expansion of mining employment due to exports also translates into additional jobs in the service sector – but not in public services. The spillover effects, nonetheless, are smaller than the ones observed in construction; and, like the latter case, these effects are exclusively observed in the group of countries with lower manufacturing.

Finally, we find that in most cases, elasticities at the national level are rather moderate if compared to regional effects. Moretti (2010) indicates that the local multipliers represent the upper-bound values for a country-wide setting; accordingly, our results are in line with this expectation. Yet, from an employment opportunities perspective, our results confirm that the mining sector provides limited room for

Table 7

Results – Services and public services employment with interaction terms for countries with higher manufacturing.

	1	2	3	4	5	6
	Δ Services employment (log)			Δ Public services employment (log)		
	GMM	GMM	GMM	GMM	GMM	GMM
Δ Mining employment – total (log)			0.073			–0.041
			–1.42			–0.54
Δ Mining employment – total (log) * <i>manufacturing</i>			–0.063			–0.042
			–0.62			–0.44
Δ Mining employment - Domestic demand (log)	0.007	0.005	–0.009	0.005	–0.004	0.016
	–0.27	–0.2	–0.26	–0.21	–0.1	–0.59
Δ Mining employment - Domestic demand (log) * <i>manufacturing</i>	–0.087	–0.029	–0.062	0.006	0.000	–0.027
	–1.25	–0.52	–1.11	–0.14	0	–0.58
Δ Mining employment - Exports (log)	0.016	0.0380 ^c	0.0402 ^a	0.011	0.033	0.016
	–0.66	–2.83	–1.9	–0.29	–1.32	–0.52
Δ Mining employment - Exports (log) * <i>manufacturing</i>	0.000	–0.035	–0.026	0.004	–0.030	0.008
	0	–0.86	–1.03	–0.11	–0.84	–0.3
Energy to non-energy mining value ratio		0.004	0.002		0.014	0.002
		–0.63	–0.54		–1.17	–0.45
Energy to non-energy mining value ratio * <i>manufacturing</i>		0.006	0.004		–0.014	0.002
		–0.63	–0.67		–1.08	–0.34
Δ GDP p.c. (log)	0.576 ^a	0.550	0.450	0.593 ^a	0.733 ^b	0.511
	–2.00	–1.29	–1.07	–1.83	–2.20	–1.15
Δ Services employment (log), t-1	–0.197	–0.102	–0.153			
	–1.58	–0.85	–1.33			
Δ Public services employment (log), t-1				0.122	0.063	–0.0181
				–0.55	–0.29	–0.07
Constant	–0.0124	–0.0163	–0.0156	–0.0222	–0.0223	–0.019
	–0.84	–1.03	–1.02	–1.31	–1.11	–0.86
Observations	440	440	440	440	440	440
Sargan p-value	0.221	0.183	0.589	0.750	0.958	0.964
Hansen p-value	0.427	0.614	0.697	0.190	0.375	0.304
AR1 p-value	0.010	0.005	0.001	0.018	0.027	0.001
AR2 p-value	0.328	0.139	0.088	0.437	0.560	0.134
Instruments (j)	37	36	37	37	36	37

t-statistics in parentheses.

^a p < 0.10.^b p < 0.05.^c p < 0.01.

generating significant positive employment spillovers at the national level.

CRedit authorship contribution statement

Beatriz Calzada Olvera: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

None.

Annex.*Sector Groups*

- **Agriculture:** All activities under A, Agriculture, forestry, and fishing. Includes:

Crop and animal production, hunting and related service activities	A01
Forestry and logging	A02
Fishing and aquaculture	A03

- **Mining:** All activities under B, Mining and quarrying
- **Manufacturing:** All activities under C, Manufacturing. Includes:

Data availability

Data will be made available on request.

Acknowledgements

I would like to sincerely express my gratitude to Prof. Neil Foster-McGregor, Prof. Bart Verspagen, and the two anonymous referees for their valuable comments and input. Their expertise and insights have greatly contributed to the improvement of this research.

Manufacture of food products, beverages, and tobacco products	C10–C12
Manufacture of textiles, wearing apparel and leather products	C13–C15
Manufacture of wood and products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	C16
Manufacture of paper and paper products	C17
Printing and reproduction of recorded media	C18
Manufacture of chemicals and chemical products	C20
Manufacture of basic pharmaceutical products and pharmaceutical preparations	C21
Manufacture of rubber and plastic products	C22
Manufacture of other non-metallic mineral products	C23
Manufacture of basic metals	C24
Manufacture of fabricated metal products, except machinery and equipment	C25
Manufacture of computer, electronic, and optical products	C26
Manufacture of electrical equipment	C27
Manufacture of machinery and equipment n.e.c.	C28
Manufacture of motor vehicles, trailers, and semi-trailers	C29
Manufacture of other transport equipment	C30
Manufacture of furniture; other manufacturing	C31_C32

- **Refining activities:** All activities under C19, Manufacture of coke and refined petroleum products
- **Construction.** All activities under F, Construction.
- **Services.** Includes:

Wholesale and retail trade and repair of motor vehicles and motorcycles	G45
Wholesale trade, except for motor vehicles and motorcycles	G46
Retail trade, except for motor vehicles and motorcycles	G47
Repair and installation of machinery and equipment	C33
Electricity, gas, steam, and air conditioning supply	D35
Water collection, treatment, and supply	E36
Sewerage; waste collection, treatment, and disposal activities; materials recovery; remediation activities and other waste management services	E37-E39
Land transport and transport via pipelines	H49
Water transport	H50
Air transport	H51
Warehousing and support activities for transportation	H52
Postal and courier activities	H53
Accommodation and food service activities	I
Publishing activities	J58
Motion picture, video, and television program production, sound recording and music publishing activities; programming and broadcasting activities	J59_J60
Telecommunications	J61
Computer programming, consultancy, and related activities; information service activities	J62_J63
Financial service activities, except insurance and pension funding	K64
Insurance, reinsurance, and pension funding, except compulsory social security	K65
Activities auxiliary to financial services and insurance activities	K66
Real estate activities	L68
Legal and accounting activities; activities of head offices; management consultancy activities	M69_M70
Architectural and engineering activities; technical training and analysis	M71
Scientific research and development	M72
Advertising and market research	M73
Other professional, scientific, and technical activities; veterinary activities	M74_M75
Administrative and support service activities	N
Other service activities	R_S

- **Public Services:** Includes

Public administration and defense; compulsory social security	O84
Education	P85
Human health and social work activities	Q

References

- Alexeev, M., Conrad, R., 2009. The elusive curse of oil. *Rev. Econ. Stat.* 91 (3), 586–598. <https://doi.org/10.1162/rest.91.3.586>.
- Aroca, P., 2001. Impacts and development in local economies based on mining: the case of the Chilean II region. *Resour. Pol.* 27 (2), 119–134. [https://doi.org/10.1016/S0301-4207\(01\)00013-7](https://doi.org/10.1016/S0301-4207(01)00013-7).
- Badeeb, R.A., Lean, H.H., Clark, J., 2017. The evolution of the natural resource curse thesis: a critical literature survey. *Resour. Pol.* 51, 123–134. <https://doi.org/10.1016/J.RESOURPOL.2016.10.015>.
- Black, D., McKinnish, T., Sanders, S., 2005. The economic impact of the coal boom and bust. *Econ. J.* 115 (503) <https://doi.org/10.1111/j.1468-0297.2005.00996.x>.
- Brunnschweiler, C.N., Bulte, E.H., 2008. The resource curse revisited and revised: a tale of paradoxes and red herrings. *J. Environ. Econ. Manag.* 55 (3), 248–264. <https://doi.org/10.1016/J.JEEM.2007.08.004>.
- Cavalcanti, T., Da Mata, D., Toscani, F., 2019. Winning the oil lottery: the impact of natural resource extraction on growth. *J. Econ. Growth* 24 (1), 79–115. <https://doi.org/10.1007/s10887-018-09161-z>.
- Cavalcanti, T., Mohaddes, K., Raissi, M., 2011. Growth, development and natural resources: New evidence using a heterogeneous panel analysis. *Q. Rev. Econ. Finance* 51 (4), 305–318. <https://doi.org/10.1016/j.qref.2011.07.007>.

- Corden, W.M., Neary, J.P., 1982. Booming sector and de-industrialisation in a small open economy. *Econ. J.* 92 (368), 825. <https://doi.org/10.2307/2232670>.
- Cosbey, A., Mann, H., Maennling, N., Toledano, P., Martin, J.G., Brauch, D., 2016. Mining a Mirage? Reassessing the shared-value paradigm in light of the technological advances in the mining sector. www.iisd.org.
- Eftimie, A., Heller, K., Strongman, J., 2009. Gender Dimensions of the Extractive Industries. http://siteresources.worldbank.org/EXTOGMC/Resources/eifd8_gender_equity.pdf.
- Ejdemo, T., Söderholm, P., 2011. Mining investment and regional development: a scenario-based assessment for Northern Sweden. *Resour. Pol.* 36 (1), 14–21. <https://doi.org/10.1016/j.resourpol.2010.08.008>.
- Fleming, D.A., Measham, T.G., 2014. Local job multipliers of mining. *Resour. Pol.* 41, 9–15. <https://doi.org/10.1016/j.resourpol.2014.02.005>.
- Fleming, D.A., Measham, T.G., 2015. Local economic impacts of an unconventional energy boom: the coal seam gas industry in Australia. *Aust. J. Agric. Resour. Econ.* 59 (1), 78–94. <https://doi.org/10.1111/1467-8489.12043>.
- Foster-McGregor, N., 2019. Global value chains and employment in asia. *Asian Dev. Rev. Stud. Asian Pac. Econ. Issues* 36 (2), 100–130. <https://www.adb.org/sites/default/files/publication/525971/asian-development-review-volume-36-number-2.pdf>.
- Foster-McGregor, N., Sartorello Spinola, D., Verspagen, B., 2018. ON the DEVELOPMENT and IMPACT of COMMODITY PRICES and CYCLES Inclusive and Sustainable Industrial Development Working Paper Series WP 8 | 2018.
- Gerelmaa, L., Kotani, K., 2016. Further investigation of natural resources and economic growth: do natural resources depress economic growth? *Resour. Pol.* 50, 312–321. <https://doi.org/10.1016/j.resourpol.2016.10.004>.
- GIZ, 2016. Linkages to the Resource Sector the Role of Companies. Government and International Development Cooperation. www.giz.de.
- James, A., 2015. The resource curse: a statistical mirage? *J. Dev. Econ.* 114, 55–63. <https://doi.org/10.1016/j.jdeveco.2014.10.006>.
- Korinek, J., 2020. The mining global value chain. In: OECD Trade Policy Papers. <https://doi.org/10.1787/2827283e-en> (Issue 235).
- Krugman, P., 1987. The narrow moving band, the Dutch disease, and the competitive consequences of Mrs. Thatcher. Notes on trade in the presence of dynamic scale economies. *J. Dev. Econ.* 27 (1–2), 41–55. [https://doi.org/10.1016/0304-3878\(87\)90005-8](https://doi.org/10.1016/0304-3878(87)90005-8).
- Lederman, D., Maloney, W.F., 2007. Trade structure and growth. In: Lederman, D., Manolev, W.F. (Eds.), *Natural Resources Neither Curse Nor Destiny*. Stanford University Press, The World Bank, pp. 15–40.
- Leeuw, P., Mtegha, H., 2018. The significance of mining backward and forward linkages in reskilling redundant mine workers in South Africa. *Resour. Pol.* <https://doi.org/10.1016/j.resourpol.2018.02.004>.
- Los, B., Timmer, M.P., de Vries, G.J., 2015. How important are exports for job growth in China? A demand side analysis. *J. Comp. Econ.* 43 (1), 19–32. <https://doi.org/10.1016/j.jce.2014.11.007>.
- Marchand, J., 2012. Local labor market impacts of energy boom-bust-boom in Western Canada. *J. Urban Econ.* 71 (1), 165–174. <https://doi.org/10.1016/J.JUE.2011.06.001>.
- Matsuyama, K., 1992. Agricultural Productivity, Comparative Advantage, and Economic Growth.
- McMahon, G., Remy, F., 2002. In: McMahon, G., Remy, F. (Eds.), *Global Mining Large Mines and Local Communities: Forging Partnerships, Building Sustainability*.
- Moretti, E., 2010. Local multipliers. *Am. Econ. Rev.* 100 (2), 373–377. <https://doi.org/10.1257/aer.100.2.373>.
- Moritz, T., Ejdemo, T., Söderholm, P., Wårell, L., 2017. The local employment impacts of mining: an econometric analysis of job multipliers in northern Sweden. *Mineral Economics* 30 (1), 53–65. <https://doi.org/10.1007/s13563-017-0103-1>.
- Morris, M., Kaplinsky, R., Kaplan, D., 2012. “One thing leads to another”—commodities, linkages and industrial development. *Resour. Pol.* 37 (4), 408–416. <https://doi.org/10.1016/J.RESOURPOL.2012.06.008>.
- Rajbhandari, I., Faggian, A., Partridge, M.D., 2022. Oil and gas boomtowns and occupations: what types of jobs are created? *Energy Econ.* 115, 106321 <https://doi.org/10.1016/j.eneco.2022.106321>.
- Roodman, D., 2009a. How to do xtabond2: an introduction to difference and system GMM in Stata. *STATA J.* 9 (Issue 1).
- Roodman, D., 2009b. A note on the theme of too many instruments. *Oxf. Bull. Econ. Stat.* 71 (1), 135–158. <https://doi.org/10.1111/j.1468-0084.2008.00542.x>.
- Sachs, J., Warner, A., 1995. Natural Resource Abundance and Economic Growth, vol. 3. National Bureau of Economic Research. <https://doi.org/10.3386/w5398>. No. 5398.
- Timmer, M.P., Dietzenbacher, E., Los, B., Stehrer, R., de Vries, G.J., 2015. An illustrated user guide to the world input–output database: The case of global automotive production. *Rev. Int. Econ.* 23 (3), 575–605. <https://doi.org/10.1111/roie.12178>.
- Torvik, R., 2001. Learning by doing and the Dutch disease. *Eur. Econ. Rev.* 45 (2), 285–306. [https://doi.org/10.1016/S0014-2921\(99\)00071-9](https://doi.org/10.1016/S0014-2921(99)00071-9).
- Van Dijk, J.J., 2018. Robustness of econometrically estimated local multipliers across different methods and data. *J. Reg. Sci.* 58 (2), 281–294. <https://doi.org/10.1111/jors.12378>.
- Wang, Z., Wei, S.-J., Yu, X., Zhu, K., 2017. Measures of Participation in Global Value Chains and Global Business Cycles, pp. 1–33. <https://doi.org/10.3386/w23222>.
- Weber, J.G., 2012. The effects of a natural gas boom on employment and income in Colorado, Texas, and Wyoming. *Energy Econ.* 34, 1580–1588. <https://doi.org/10.1016/j.eneco.2011.11.013>.
- Weber, J.G., 2014. A decade of natural gas development: the makings of a resource curse? *Resour. Energy Econ.* 37, 168–183. <https://doi.org/10.1016/j.reseneeco.2013.11.013>.